

MATH 350-2 Advanced Calculus

W.R. Casper

Department of Mathematics
California State University Fullerton

November 12, 2025

Outline

- 1 Real Analysis Lecture 23
 - Differentiation
 - Derivatives and Optima
 - Mean Value Theorem

Outline

- 1 Real Analysis Lecture 23
 - Differentiation
 - Derivatives and Optima
 - Mean Value Theorem

Differentiability

Definition

A real-valued function $f(x)$ defined on an open interval (a, b) is **differentiable** at $c \in (a, b)$ if

$$\lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c} \text{ exists.}$$

In this case the limit is called the derivative of f at c , and denote by $f'(c)$.

Chord function

Theorem (Helper Theorem)

Let f be a real-valued function on (a, b) . Then f is differentiable at $c \in (a, b)$ if and only if there exists a function f^ on (a, b) which is continuous at c and satisfies*

$$f(x) - f(c) = (x - c)f^*(x).$$

Chord function

Theorem (Helper Theorem)

Let f be a real-valued function on (a, b) . Then f is differentiable at $c \in (a, b)$ if and only if there exists a function f^ on (a, b) which is continuous at c and satisfies*

$$f(x) - f(c) = (x - c)f^*(x).$$

The function $f^*(x)$ is the **chord slope function**.

Chord function

Proof.

Chord function

Proof.

Suppose $f(x)$ is differentiable at c .

Chord function

Proof.

Suppose $f(x)$ is differentiable at c . Define

$$f^*(x) = \begin{cases} \frac{f(x)-f(c)}{x-c}, & x \neq c \\ f'(c), & x = c \end{cases}$$

Chord function

Proof.

Suppose $f(x)$ is differentiable at c . Define

$$f^*(x) = \begin{cases} \frac{f(x)-f(c)}{x-c}, & x \neq c \\ f'(c), & x = c \end{cases}$$

Then $\lim_{x \rightarrow c} f^*(x) = f'(c)$, so f^* is continuous at c .

Chord function

Proof.

Suppose $f(x)$ is differentiable at c . Define

$$f^*(x) = \begin{cases} \frac{f(x)-f(c)}{x-c}, & x \neq c \\ f'(c), & x = c \end{cases}$$

Then $\lim_{x \rightarrow c} f^*(x) = f'(c)$, so f^* is continuous at c .
Conversely, suppose f^* exists and is continuous at c .

Chord function

Proof.

Suppose $f(x)$ is differentiable at c . Define

$$f^*(x) = \begin{cases} \frac{f(x)-f(c)}{x-c}, & x \neq c \\ f'(c), & x = c \end{cases}$$

Then $\lim_{x \rightarrow c} f^*(x) = f'(c)$, so f^* is continuous at c .
Conversely, suppose f^* exists and is continuous at c .
Then

$$\lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c} = \lim_{x \rightarrow c} f^*(x) = f^*(c).$$

Chord function

Proof.

Suppose $f(x)$ is differentiable at c . Define

$$f^*(x) = \begin{cases} \frac{f(x)-f(c)}{x-c}, & x \neq c \\ f'(c), & x = c \end{cases}$$

Then $\lim_{x \rightarrow c} f^*(x) = f'(c)$, so f^* is continuous at c .
Conversely, suppose f^* exists and is continuous at c .
Then

$$\lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c} = \lim_{x \rightarrow c} f^*(x) = f^*(c).$$

In particular f is differentiable at c and $f'(x) = f^*(c)$. □

Differentiable implies continuous

Theorem

Suppose $f(x)$ is a real valued function on (a, b) and is differentiable at $c \in (a, b)$.

Then f is continuous at c .

Differentiable implies continuous

Theorem

*Suppose $f(x)$ is a real valued function on (a, b) and is differentiable at $c \in (a, b)$.
Then f is continuous at c .*

Proof.

Differentiable implies continuous

Theorem

Suppose $f(x)$ is a real valued function on (a, b) and is differentiable at $c \in (a, b)$.

Then f is continuous at c .

Proof.

By the Helper Theorem, there exists a function f^* on (a, b) which is continuous at c with

$$f(x) = (x - c)f^*(x).$$

Differentiable implies continuous

Theorem

Suppose $f(x)$ is a real valued function on (a, b) and is differentiable at $c \in (a, b)$.

Then f is continuous at c .

Proof.

By the Helper Theorem, there exists a function f^* on (a, b) which is continuous at c with

$$f(x) = (x - c)f^*(x).$$

Since $(x - c)$ and f^* are both continuous at c , so is $f(x)$. □

Additivity of derivatives

Theorem

Let f, g be real functions on (a, b) which are differentiable at $c \in (a, b)$.

$$(f \pm g)'(c) = f'(c) \pm g'(c)$$

Additivity of derivatives

Proof.

Additivity of derivatives

Proof.

By the Helper Theorem

$$f(x) - f(c) = (x - c)f^*(x), \quad g(x) - g(c) = (x - c)g^*(x)$$

Additivity of derivatives

Proof.

By the Helper Theorem

$$f(x) - f(c) = (x - c)f^*(x), \quad g(x) - g(c) = (x - c)g^*(x)$$

Therefore

$$(f \pm g)(x) - (f \pm g)(c) = (x - c)(f^*(x) \pm g^*(x)).$$

Additivity of derivatives

Proof.

By the Helper Theorem

$$f(x) - f(c) = (x - c)f^*(x), \quad g(x) - g(c) = (x - c)g^*(x)$$

Therefore

$$(f \pm g)(x) - (f \pm g)(c) = (x - c)(f^*(x) \pm g^*(x)).$$

Since f^* and g^* are continuous at c , so is $f^* \pm g^*$.

Additivity of derivatives

Proof.

By the Helper Theorem

$$f(x) - f(c) = (x - c)f^*(x), \quad g(x) - g(c) = (x - c)g^*(x)$$

Therefore

$$(f \pm g)(x) - (f \pm g)(c) = (x - c)(f^*(x) \pm g^*(x)).$$

Since f^* and g^* are continuous at c , so is $f^* \pm g^*$.

The Helper Theorem implies that $g \pm f$ is differentiable and

$$(f \pm g)'(c) = f'(c) \pm g'(c)$$



Product rule

Theorem

Let f, g be real functions on (a, b) which are differentiable at $c \in (a, b)$.

$$(fg)'(c) = f'(c)g(c) + f(c)g'(c)$$

Product rule

Proof.

Product rule

Proof.

By the Helper Theorem

$$f(x) - f(c) = (x - c)f^*(x), \quad g(x) - g(c) = (x - c)g^*(x)$$

Product rule

Proof.

By the Helper Theorem

$$f(x) - f(c) = (x - c)f^*(x), \quad g(x) - g(c) = (x - c)g^*(x)$$

Therefore

$$\begin{aligned}(fg)(x) - (fg)(c) &= f(x)g(x) - f(c)g(c) \\&= f(x)g(x) - f(x)g(c) + f(x)g(c) - f(c)g(c) \\&= f(x)(g(x) - g(c)) + (f(x) - f(c))g(c) \\&= (x - c)[f(x)g^*(x) + f^*(x)g(c)]\end{aligned}$$

Product rule

Proof.

By the Helper Theorem

$$f(x) - f(c) = (x - c)f^*(x), \quad g(x) - g(c) = (x - c)g^*(x)$$

Therefore

$$\begin{aligned}(fg)(x) - (fg)(c) &= f(x)g(x) - f(c)g(c) \\&= f(x)g(x) - f(x)g(c) + f(x)g(c) - f(c)g(c) \\&= f(x)(g(x) - g(c)) + (f(x) - f(c))g(c) \\&= (x - c)[f(x)g^*(x) + f^*(x)g(c)]\end{aligned}$$

Helper Theorem says:

$$(fg)'(c) = f(c)g^*(c) + f^*(c)g(c) = f(c)g'(c) + f'(c)g(c).$$

Quotient rule

Theorem

Let f, g be real functions on (a, b) which are differentiable at $c \in (a, b)$ with $g(c) \neq 0$.

$$(f/g)'(c) = \frac{f'(c)g(c) - f(c)g'(c)}{g(c)^2}$$

Quotient rule

By the Helper Theorem

$$f(x) - f(c) = (x - c)f^*(x), \quad g(x) - g(c) = (x - c)g^*(x)$$

Quotient rule

By the Helper Theorem

$$f(x) - f(c) = (x - c)f^*(x), \quad g(x) - g(c) = (x - c)g^*(x)$$

$$\begin{aligned} (f/g)(x) - (f/g)(c) &= \frac{f(x)g(c) - f(c)g(x)}{g(x)g(c)} \\ &= \frac{f(x)g(c) - f(c)g(c) + f(c)g(c) - f(c)g(x)}{g(x)g(c)} \\ &= \frac{(f(x) - f(c))g(c) - f(c)(g(x) - g(c))}{g(x)g(c)} \\ &= (x - c) \frac{f^*(x)g(c) - f(c)g^*(x)}{g(x)g(c)} \end{aligned}$$

Quotient rule

By the Helper Theorem

$$f(x) - f(c) = (x - c)f^*(x), \quad g(x) - g(c) = (x - c)g^*(x)$$

$$\begin{aligned} (f/g)(x) - (f/g)(c) &= \frac{f(x)g(c) - f(c)g(x)}{g(x)g(c)} \\ &= \frac{f(x)g(c) - f(c)g(c) + f(c)g(c) - f(c)g(x)}{g(x)g(c)} \\ &= \frac{(f(x) - f(c))g(c) - f(c)(g(x) - g(c))}{g(x)g(c)} \\ &= (x - c) \frac{f^*(x)g(c) - f(c)g^*(x)}{g(x)g(c)} \end{aligned}$$

Helper Theorem says:

$$(f/g)'(c) = \frac{f^*(c)g(c) - f(c)g^*(c)}{g(c)^2} = \frac{f'(c)g(c) - f(c)g'(c)}{g(c)^2}.$$

Chain rule

Theorem

Let f be a real function on an interval $I = (a, b)$, and let g be a real function on $f(I)$, and consider the composition

$$h = g \circ f : (a, b) \rightarrow \mathbb{R}.$$

Suppose $c \in (a, b)$ and $f(c)$ is an interior point of $f(I)$. If f is differentiable at c and g is differentiable at $f(c)$, then h is differentiable at c and

$$h'(c) = (g \circ f)'(c) = g'(f(c))f'(c).$$

Outline

- 1 Real Analysis Lecture 23
 - Differentiation
 - Derivatives and Optima
 - Mean Value Theorem

Local maxima and minima

Definition

Let (M, d) be a metric space and f be a real-valued function on a subset $S \subseteq M$.

Local maxima and minima

Definition

Let (M, d) be a metric space and f be a real-valued function on a subset $S \subseteq M$. Then f has an **absolute (or global) maximum at $c \in S$** if

$$f(x) \leq f(c) \quad \text{for all } x \in S.$$

Local maxima and minima

Definition

Let (M, d) be a metric space and f be a real-valued function on a subset $S \subseteq M$. Then f has an **absolute (or global) maximum at $c \in S$** if

$$f(x) \leq f(c) \quad \text{for all } x \in S.$$

It has **local maximum at $a \in S$** if there exists an $r > 0$ such that

$$f(x) \leq f(c) \quad \text{for all } x \in B_M(c; r) \cap S.$$

Local maxima and minima

Definition

Let (M, d) be a metric space and f be a real-valued function on a subset $S \subseteq M$. Then f has an **absolute (or global) maximum at $c \in S$** if

$$f(x) \leq f(c) \quad \text{for all } x \in S.$$

It has **local maximum at $a \in S$** if there exists an $r > 0$ such that

$$f(x) \leq f(c) \quad \text{for all } x \in B_M(c; r) \cap S.$$

Absolute and local minima are defined similarly.

Critical points

Definition

Let f be a real-valued function on an open interval (a, b) .

Critical points

Definition

Let f be a real-valued function on an open interval (a, b) . A point $c \in (a, b)$ is called a **critical point** of f if either f is not differentiable at c or else $f'(c) = 0$.

Critical points

Definition

Let f be a real-valued function on an open interval (a, b) . A point $c \in (a, b)$ is called a **critical point** of f if either f is not differentiable at c or else $f'(c) = 0$.

Theorem

Let f be a real-valued function on an open interval (a, b) and suppose that f has a local maximum or local minimum at $c \in (a, b)$. Then c is a critical point of f .

Critical points

Proof.

Critical points

Proof.

We will prove this in the case c is a local maximum. The proof in the case of a local minimum is similar.

Critical points

Proof.

We will prove this in the case c is a local maximum. The proof in the case of a local minimum is similar.

If f is not differentiable at c , then we're done!

Critical points

Proof.

We will prove this in the case c is a local maximum. The proof in the case of a local minimum is similar.

If f is not differentiable at c , then we're done!

Therefore assume that it is differentiable.

Critical points

Proof.

We will prove this in the case c is a local maximum. The proof in the case of a local minimum is similar.

If f is not differentiable at c , then we're done!

Therefore assume that it is differentiable.

By definition, there exists $r > 0$ such that

$$f(x) \leq f(c) \quad \text{for all } x \in (c - r, c + r) \cap (a, b).$$

Critical points

Proof.

We will prove this in the case c is a local maximum. The proof in the case of a local minimum is similar.

If f is not differentiable at c , then we're done!

Therefore assume that it is differentiable.

By definition, there exists $r > 0$ such that

$$f(x) \leq f(c) \quad \text{for all } x \in (c - r, c + r) \cap (a, b).$$

To obtain a contradiction, suppose that $f'(c) \neq 0$.

Critical points

Proof.

We will prove this in the case c is a local maximum. The proof in the case of a local minimum is similar.

If f is not differentiable at c , then we're done!

Therefore assume that it is differentiable.

By definition, there exists $r > 0$ such that

$$f(x) \leq f(c) \quad \text{for all } x \in (c - r, c + r) \cap (a, b).$$

To obtain a contradiction, suppose that $f'(c) \neq 0$.

Then by the Helper Theorem, there is a function $f^*(x)$ defined on (a, b) and continuous at $c \in (a, b)$ with

$$f(x) - f(c) = (x - c)f^*(x).$$

Critical points

Proof.

Moreover, $f^*(c) = f'(c) \neq 0$.

Critical points

Proof.

Moreover, $f^*(c) = f'(c) \neq 0$.

Since f is continuous, we may choose $\delta > 0$ such that

$$|x - c| < \delta \Rightarrow |f^*(x) - f^*(c)| < |f^*(c)|/2.$$

Critical points

Proof.

Moreover, $f^*(c) = f'(c) \neq 0$.

Since f is continuous, we may choose $\delta > 0$ such that

$$|x - c| < \delta \Rightarrow |f^*(x) - f^*(c)| < |f^*(c)|/2.$$

Moreover, we may assume that $\delta < r$.

Critical points

Proof.

Moreover, $f^*(c) = f'(c) \neq 0$.

Since f is continuous, we may choose $\delta > 0$ such that

$$|x - c| < \delta \Rightarrow |f^*(x) - f^*(c)| < |f^*(c)|/2.$$

Moreover, we may assume that $\delta < r$.

It follows that $f^*(x) > |f^*(c)|/2$ for all $x \in (a, b)$ with $|x - c| < \delta$.

Critical points

Proof.

Moreover, $f^*(c) = f'(c) \neq 0$.

Since f is continuous, we may choose $\delta > 0$ such that

$$|x - c| < \delta \Rightarrow |f^*(x) - f^*(c)| < |f^*(c)|/2.$$

Moreover, we may assume that $\delta < r$.

It follows that $f^*(x) > |f^*(c)|/2$ for all $x \in (a, b)$ with $|x - c| < \delta$.

However,

$$f(c \pm \delta/2) = f(c) \pm \frac{1}{2}\delta f^*(c \pm \delta/2),$$

one of these will be larger than $f(c)$.

Critical points

Proof.

Moreover, $f^*(c) = f'(c) \neq 0$.

Since f is continuous, we may choose $\delta > 0$ such that

$$|x - c| < \delta \Rightarrow |f^*(x) - f^*(c)| < |f^*(c)|/2.$$

Moreover, we may assume that $\delta < r$.

It follows that $f^*(x) > |f^*(c)|/2$ for all $x \in (a, b)$ with $|x - c| < \delta$.

However,

$$f(c \pm \delta/2) = f(c) \pm \frac{1}{2}\delta f^*(c \pm \delta/2),$$

one of these will be larger than $f(c)$. Contradiction! □

Outline

- 1 Real Analysis Lecture 23
 - Differentiation
 - Derivatives and Optima
 - Mean Value Theorem

Rolle's Theorem

Theorem

Rolle's Theorem Suppose that $f(x)$ is a real-valued function on $[a, b]$ which is

Rolle's Theorem

Theorem

Rolle's Theorem Suppose that $f(x)$ is a real-valued function on $[a, b]$ which is

- *continuous on $[a, b]$;*

Rolle's Theorem

Theorem

Rolle's Theorem Suppose that $f(x)$ is a real-valued function on $[a, b]$ which is

- *continuous on $[a, b]$;*
- *differentiable at each point in (a, b) ;*

Rolle's Theorem

Theorem

Rolle's Theorem Suppose that $f(x)$ is a real-valued function on $[a, b]$ which is

- *continuous on $[a, b]$;*
- *differentiable at each point in (a, b) ;*
- *and which satisfies $f(a) = f(b)$.*

Rolle's Theorem

Theorem

Rolle's Theorem Suppose that $f(x)$ is a real-valued function on $[a, b]$ which is

- *continuous on $[a, b]$;*
- *differentiable at each point in (a, b) ;*
- *and which satisfies $f(a) = f(b)$.*

Then there exists $c \in (a, b)$ with $f'(c) = 0$.

Rolle's Theorem

Proof.

Suppose otherwise.

Rolle's Theorem

Proof.

Suppose otherwise.

By the Extreme Value Theorem, $f(x)$ has an absolute maximum on $[a, b]$, as well as an absolute minimum on $[a, b]$.

Rolle's Theorem

Proof.

Suppose otherwise.

By the Extreme Value Theorem, $f(x)$ has an absolute maximum on $[a, b]$, as well as an absolute minimum on $[a, b]$. If both of these occur at the endpoints, then the absolute max is the absolute min, forcing f to be constant.

Rolle's Theorem

Proof.

Suppose otherwise.

By the Extreme Value Theorem, $f(x)$ has an absolute maximum on $[a, b]$, as well as an absolute minimum on $[a, b]$.

If both of these occur at the endpoints, then the absolute max is the absolute min, forcing f to be constant.

Then $f'(c) = 0$ for every $c \in (a, b)$.

Rolle's Theorem

Proof.

Suppose otherwise.

By the Extreme Value Theorem, $f(x)$ has an absolute maximum on $[a, b]$, as well as an absolute minimum on $[a, b]$. If both of these occur at the endpoints, then the absolute max is the absolute min, forcing f to be constant.

Then $f'(c) = 0$ for every $c \in (a, b)$.

Otherwise at least one of the points happens inside, so $f(x)$ has a local max or local min at a value c inside (a, b) .

Rolle's Theorem

Proof.

Suppose otherwise.

By the Extreme Value Theorem, $f(x)$ has an absolute maximum on $[a, b]$, as well as an absolute minimum on $[a, b]$. If both of these occur at the endpoints, then the absolute max is the absolute min, forcing f to be constant.

Then $f'(c) = 0$ for every $c \in (a, b)$.

Otherwise at least one of the points happens inside, so $f(x)$ has a local max or local min at a value c inside (a, b) .

The point c is a critical point, and since f is differentiable at c , we must have $f'(c) = 0$. □

Mean Value Theorem

Theorem

Rolle's Theorem Suppose that $f(x)$ is a real-valued function on $[a, b]$ which is

Mean Value Theorem

Theorem

Rolle's Theorem Suppose that $f(x)$ is a real-valued function on $[a, b]$ which is

- continuous on $[a, b]$;

Mean Value Theorem

Theorem

Rolle's Theorem Suppose that $f(x)$ is a real-valued function on $[a, b]$ which is

- *continuous on $[a, b]$;*
- *and differentiable at each point in (a, b) .*

Mean Value Theorem

Theorem

Rolle's Theorem Suppose that $f(x)$ is a real-valued function on $[a, b]$ which is

- *continuous on $[a, b]$;*
- *and differentiable at each point in (a, b) .*

Then there exists $c \in (a, b)$ with

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

Mean Value Theorem

Proof.

Consider the function

$$g(x) = f(x) - \left(\frac{f(b) - f(a)}{b - a} \right) (x - a) + f(a).$$

Mean Value Theorem

Proof.

Consider the function

$$g(x) = f(x) - \left(\frac{f(b) - f(a)}{b - a} \right) (x - a) + f(a).$$

Then $g(x)$ is

Mean Value Theorem

Proof.

Consider the function

$$g(x) = f(x) - \left(\frac{f(b) - f(a)}{b - a} \right) (x - a) + f(a).$$

Then $g(x)$ is

- continuous on $[a, b]$;

Mean Value Theorem

Proof.

Consider the function

$$g(x) = f(x) - \left(\frac{f(b) - f(a)}{b - a} \right) (x - a) + f(a).$$

Then $g(x)$ is

- continuous on $[a, b]$;
- differentiable at each point in (a, b) ;

Mean Value Theorem

Proof.

Consider the function

$$g(x) = f(x) - \left(\frac{f(b) - f(a)}{b - a} \right) (x - a) + f(a).$$

Then $g(x)$ is

- continuous on $[a, b]$;
- differentiable at each point in (a, b) ;
- and satisfies $g(a) = g(b)$.

Mean Value Theorem

Proof.

Consider the function

$$g(x) = f(x) - \left(\frac{f(b) - f(a)}{b - a} \right) (x - a) + f(a).$$

Then $g(x)$ is

- continuous on $[a, b]$;
- differentiable at each point in (a, b) ;
- and satisfies $g(a) = g(b)$.

By Rolle's Theorem, there exists $c \in (a, b)$ with

$$0 = g'(c) = f'(c) - \frac{f(b) - f(a)}{b - a}.$$